

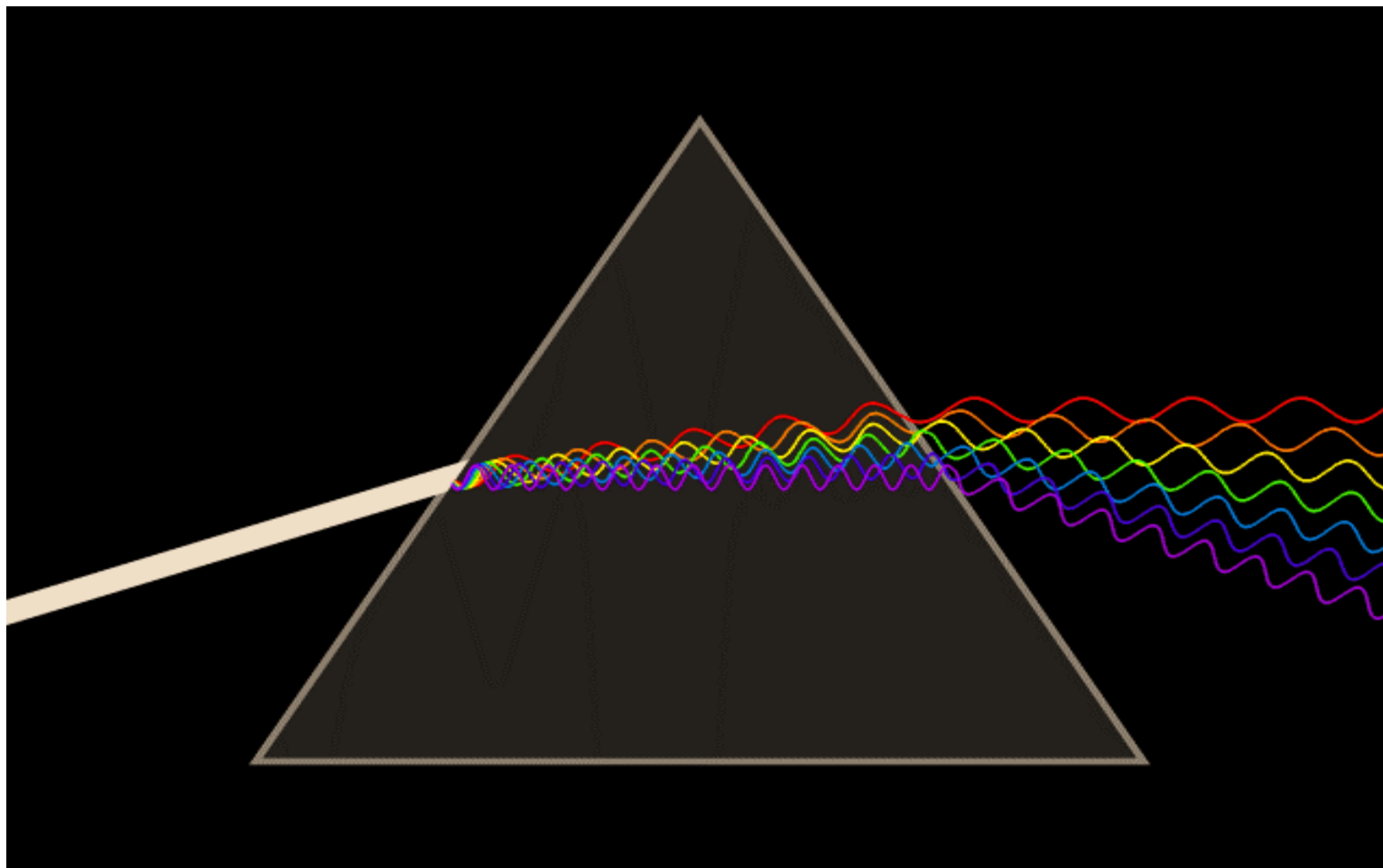


# Fundamentals of EM Waves

112-BOLCEW06

# Concrete Experience

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Q: Why should you care about electromagnetism?

# Application

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- A: The Electromagnetic Force is one of the 4 fundamental forces in the universe and describes most force interactions in your day-to-day life.
- A: Radio waves and Light waves are electromagnetic waves.
- A: Modern-day warfare depends on electromagnetic waves.
- A: The fundamentals of Electromagnetism is the cornerstone of Electromagnetic Warfare.

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# Learning Step Activities

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- Define Electromagnetic (EM) waves
- Recognize the characteristics of an EM wave
- Define magnetic, electric, electromagnetic, and radiation fields
- Calculate wave measurements
- Define how EM waves are polarized
- Define polarization
- Describe different EM wave interactions

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# Terminal Learning Objective

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The student will identify radio wave fundamentals and propagation , in a classroom environment, given a networked computer workstation and appropriate references by demonstrating understanding on definitions and facts associated with radio waves to support the application of EW effects critical to the unit's mission, in accordance with TC 9-64, EW Fundamentals Student Guide Vol. I, and NEETS, while achieving a minimum of 70% on all Practical Exercises (PE) and end of module exam.

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# What are Electromagnetic Waves?

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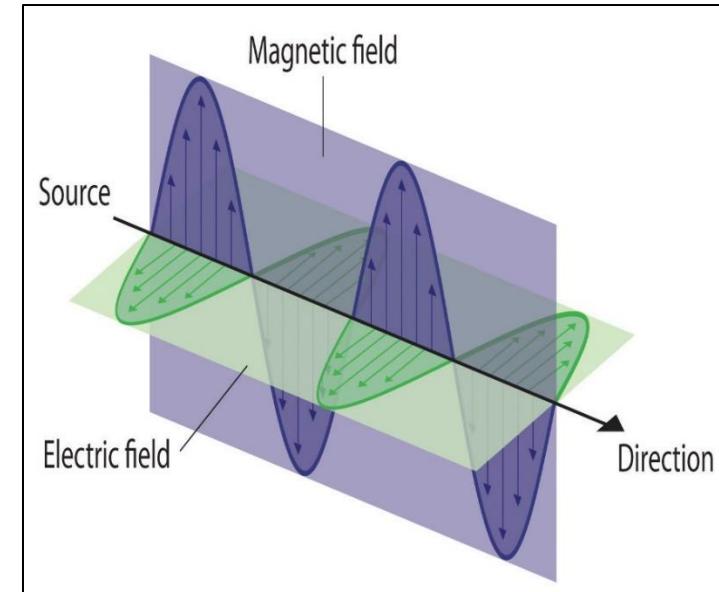
## Electromagnetic (EM) Waves are:

- A self-propagating energy wave comprised of an **Electric** and **Magnetic** field that moves at the speed of light

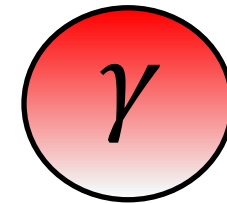
## Fundamental of an EM Wave:

- Moves at a fixed speed (speed of light)
- Is comprised of an **Electric** and **Magnetic** Field
- Has no Mass (all energy/no matter)
- Has a net zero charge
- Is self-propagating (requires no medium for transmission)
- Acts as both a particle (photon) and a wave

## EM Wave



## Photon



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# Anatomy of a classical Wave

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**Crest:** Highest Point of the Wave

**Trough:** Lowest point of the Wave

**Cycle:** one complete phase of the wave from 0-2Pi

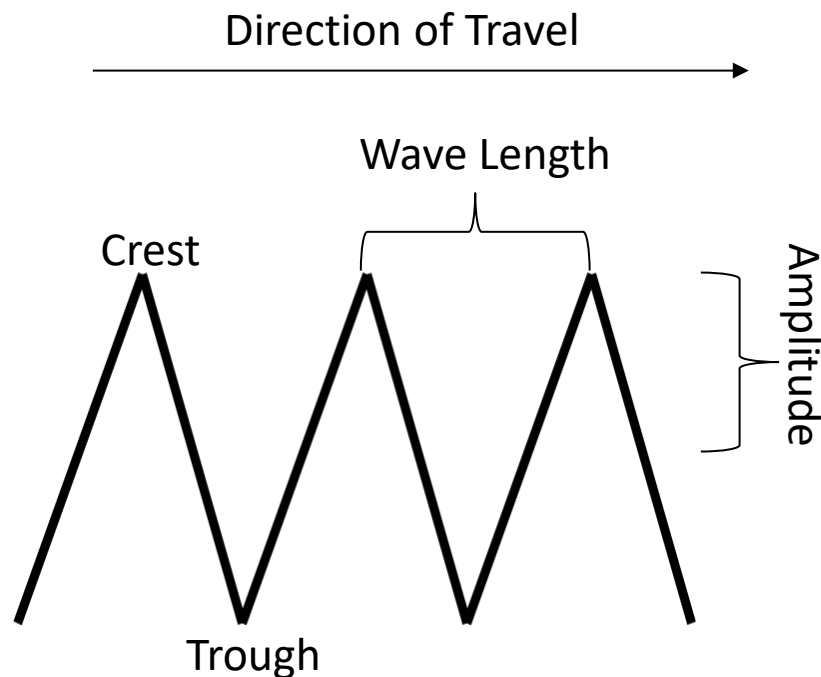
**Period:** Time taken to complete one cycle

**Wavelength:** Distance between any two matching points separated by one cycle of the wave i.e. (crest to crest)

**Amplitude:** Vertical distance between Crest and Trough

**Frequency:** Number of complete cycles during a measured time or the inverse of the period

**Speed:** Wavelength multiplied by the frequency



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# Wave Formulas

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- Hertz = the unit of measurement for frequency = cycles/time (in seconds)
- Frequency = F (measured in Hertz)= the inverse of the period (T) or the velocity (V) divided by wavelength ( $\lambda$ )

$$F = \frac{1}{T}$$

$$F = \frac{V}{\lambda}$$

- Velocity = V (measured in meter/seconds)= Wavelength ( $\lambda$ ) multiplied by the Frequency (F)

$$V = \lambda F$$

- Wavelength =  $\lambda$  (measured in meters)= velocity (V) divided by frequency (F)

$$\lambda = \frac{V}{F}$$



# Anatomy of an EM Wave

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**Electric Field Wave:** An oscillating field of zero net charge that creates a Magnetic field perpendicular to it

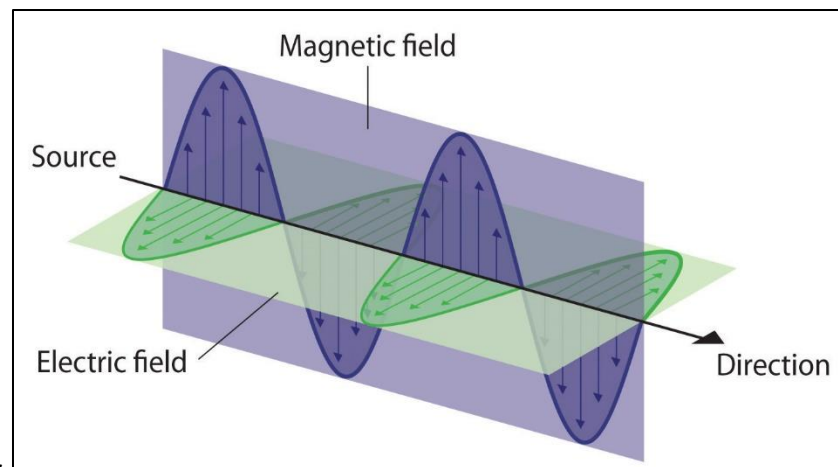
**Magnetic Field Wave:** An oscillating Magnetic field that creates an electric field perpendicular to it

**Speed:** Speed of light (Always!)

**Frequency:** Number of complete cycles during a measured time

**Wavelength:** Found by dividing the speed of light by the frequency of the wave

**Amplitude:** Proportional to square root of the wave's power divided by its area (intensity)



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# EM Wave Formulas

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- Velocity = Speed of light =  $c = 300,000,000$  meters/second
- Frequency =  $F$  (measured in Hertz) = Speed of light ( $c$ ) divided by wavelength ( $\lambda$ )

$$F = \frac{c}{\lambda}$$

- Wavelength =  $\lambda$  (measured in meters) = velocity ( $V$ ) divided by frequency ( $F$ )

$$\lambda = \frac{c}{F}$$

- The wavelength and frequency are inversely proportional at a fixed ratio (speed of light). Changing the frequency requires a change of wavelength!

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# Frequency and Wavelength Relationships

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## Frequency versus wavelength

- These are “inversely” or opposite proportions
- High Frequency: Short Wavelength; Low Frequency: Long Wavelength



## FREQUENCY

3 kHz      30 kHz      300 kHz      3 MHz      30 MHz      300 MHz      3 GHz      30 GHz      300 GHz



100 km      10 km      1 km      100 m      10 m      1 m      10 cm      1 cm      1 mm



## WAVELENGTH

|              |                 |              |                |              |                  |
|--------------|-----------------|--------------|----------------|--------------|------------------|
| AM Radio     | 535 to 1635 KHz | FM Radio     | 88 to 108 MHz  | Pagers       | 926 to 940 Mhz   |
| SW Radio     | 5.9 to 26.1 MHz | Military UHF | 225 to 400 MHz | ATC Radar    | 960 to 1215 MHz  |
| Military VHF | 20 to 80 MHz    | TV Ch 7-13   | 174 to 216 MHz | GPS          | 1227 to 1575 MHz |
| TV Ch 2-8    | 54 to 88 MHz    | TV Ch 14-83  | 470 to 806 Mhz | EW Radar     | 300 to 3000 MHz  |
|              |                 | Cell Phone   | 806 to 890 MHz | TA/TT Radars | 3 to 30 Ghz      |

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# Review of International System of Units (SI)

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- The basic system of measurement used in scientific measurement
- Used in the measurement of radio wave frequency and wavelength
- A prefix may be added to units to add to the multiple, all at the powers of ten

Example:

| SI-Prefixes   |           |           |              |              |             |             |             |             |            |            |
|---------------|-----------|-----------|--------------|--------------|-------------|-------------|-------------|-------------|------------|------------|
| <b>Name</b>   | Yotta     | Zetta     | Exa          | Peta         | Tera        | <b>Giga</b> | <b>Mega</b> | <b>Kilo</b> | Hecto      | Deca       |
| <b>Symbol</b> | Y         | Z         | E            | P            | T           | <b>G</b>    | <b>M</b>    | <b>k</b>    | h          | da         |
| <b>Factor</b> | $10^{24}$ | $10^{21}$ | $10^{18}$    | $10^{15}$    | $10^{12}$   | $10^9$      | $10^6$      | $10^3$      | $10^2$     | $10^1$     |
| SI-Prefixes   |           |           |              |              |             |             |             |             |            |            |
| <b>Name</b>   | deci      | centi     | <b>milli</b> | <b>micro</b> | <b>nano</b> | <b>pico</b> | femto       | atto        | zepto      | yocto      |
| <b>Symbol</b> | d         | c         | <b>m</b>     | <b>μ</b>     | <b>n</b>    | <b>p</b>    | f           | a           | z          | y          |
| <b>Factor</b> | $10^{-1}$ | $10^{-2}$ | $10^{-3}$    | $10^{-6}$    | $10^{-9}$   | $10^{-12}$  | $10^{-15}$  | $10^{-18}$  | $10^{-21}$ | $10^{-24}$ |

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# Example Problems

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- Find the frequency of an EM wave with a Wavelength of 1 meter

$$F = c/\lambda$$

$$\frac{(3 \times 10^8 \text{ m/s})}{1 \text{ m}} = 300 \text{ MHz}$$

- Convert 2000KHz into a MHz measurement

$$2000 \text{ KHz} = 2 \text{ MHz}$$



# Electric and Magnetic Fields

# Electric Fields

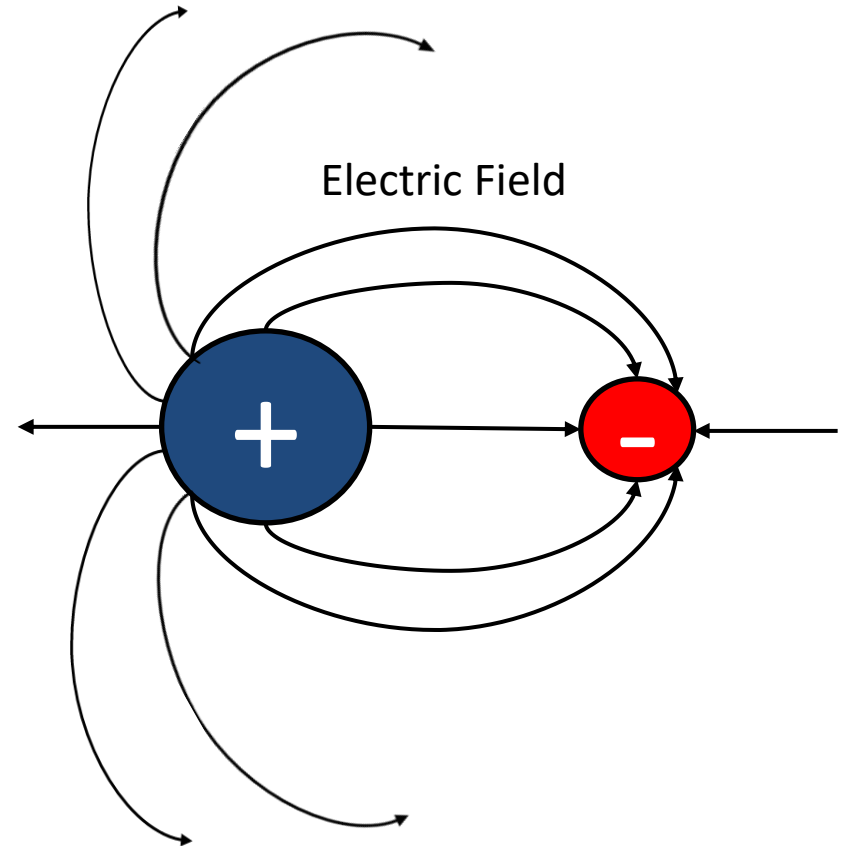
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**Charge:** A fundamental Property of matter that can be positive, negative or neutral. Charges can exist as a single point.

**Electric Force:** A force experienced by and exchanged between charged particles.

**Electric Field:** An energy field around charged particles measured as the amount of electric force per unit of charge (volts).

**Electric Field Lines:** lines that describe the orientation and direction of the field (moves from positive to negative).







# Magnetic Fields

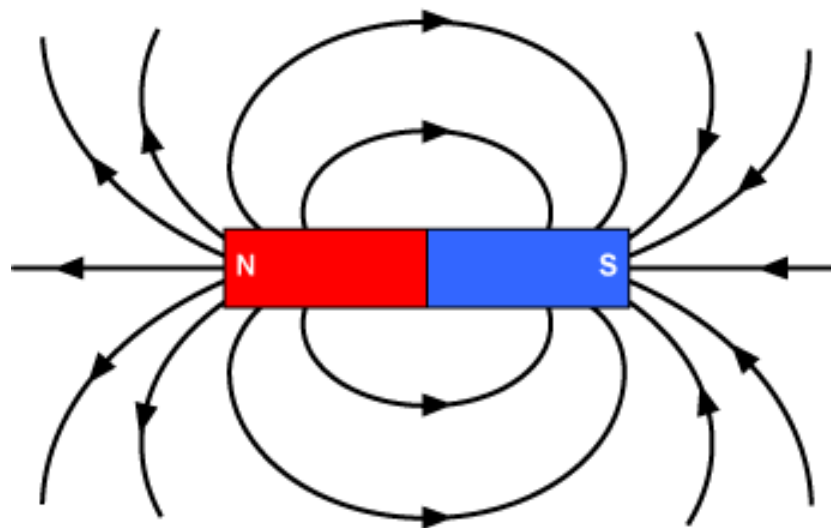
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**Magnetic Fields:** Energy field created in the presence of a changing or moving Electric Field (Does not exist on its own).

**Magnetic Field Lines:** Lines that describe the orientation and direction of the field (moves from North to South).

**Magnetic Poles:** Alignment of the magnetic field in a particular area designated as either north or south. Can not exist as mono pole (i.e., a north pole requires a south pole and vice versa).

**Magnetic Force:** A perceived force caused by the movement or change of charge.

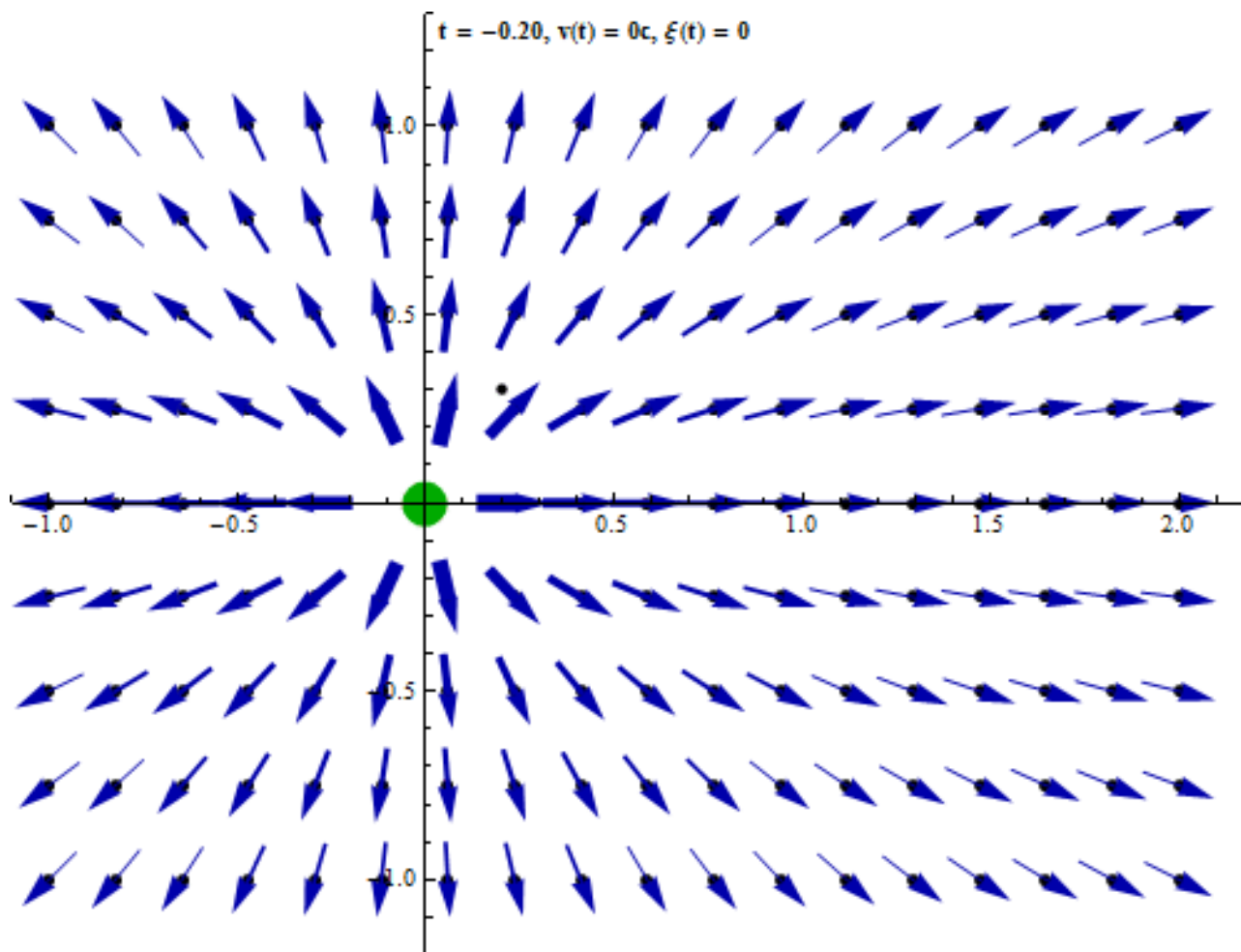


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# Fields in Motion

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# Centuries of Discovery

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Late 1500's: William Gilbert is the first to conduct investigations into magnetism



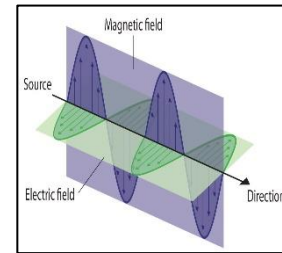
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1752: Benjamin Fr discovers Electric charge



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1785: Charles Augustin Coulomb discovers the electric force



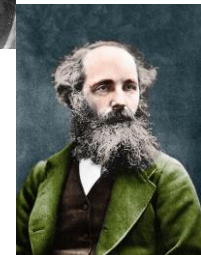
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1820: Hans Christian Oersted discovers relationship between electricity and magnetism



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1830: Michael Faraday discovers magnetic fields generate current (induction)



1861: James Maxwell unifies electricity and magnetism

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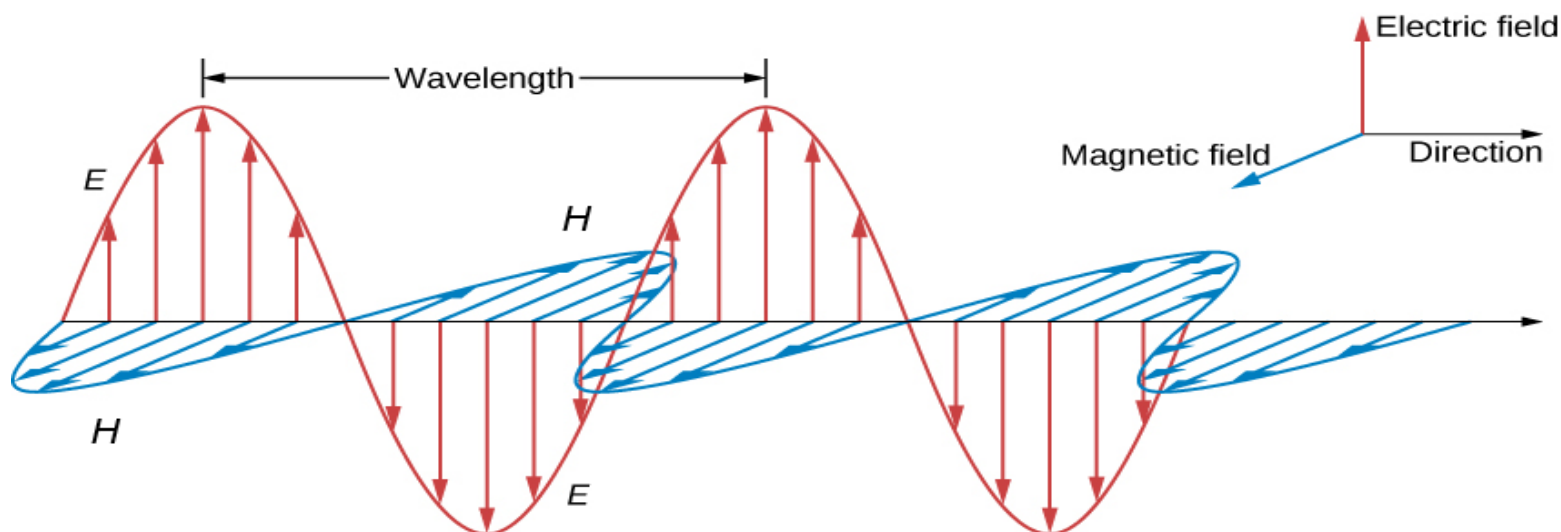


# What is the Electromagnetic Force?

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**Electromagnetic Force:** The force produced by a changing, interaction, or movement of electric and magnetic fields. It is perpendicular to both the Electric force and Magnetic force.

**Electromagnetic Field:** An energy field comprised of an electric and magnetic field. Occurs only in the presence of a changing or moving charge.



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# Questions?



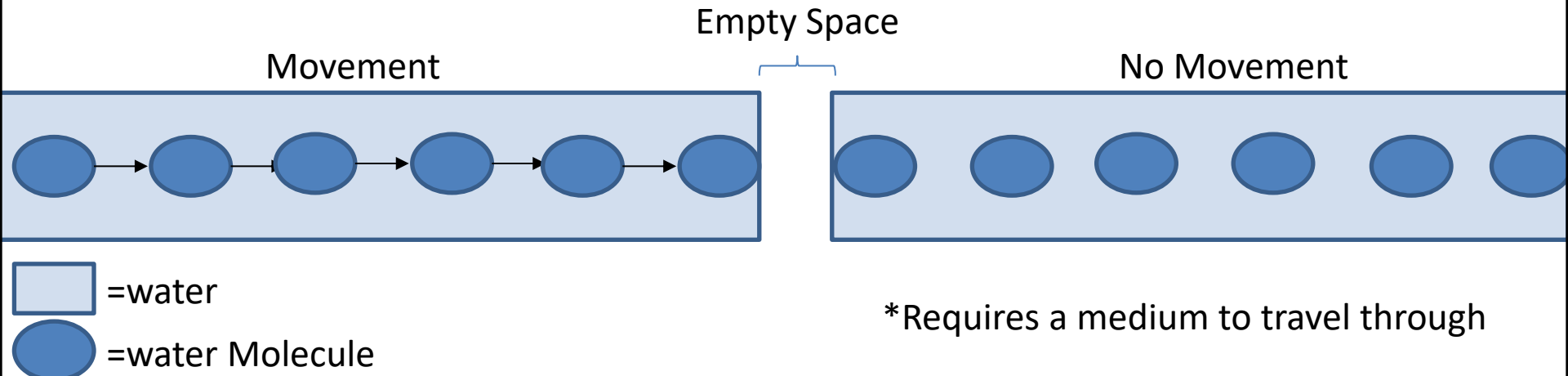
# Wave Propagation Basics



# How Normal Waves Move

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## A Wave in Water



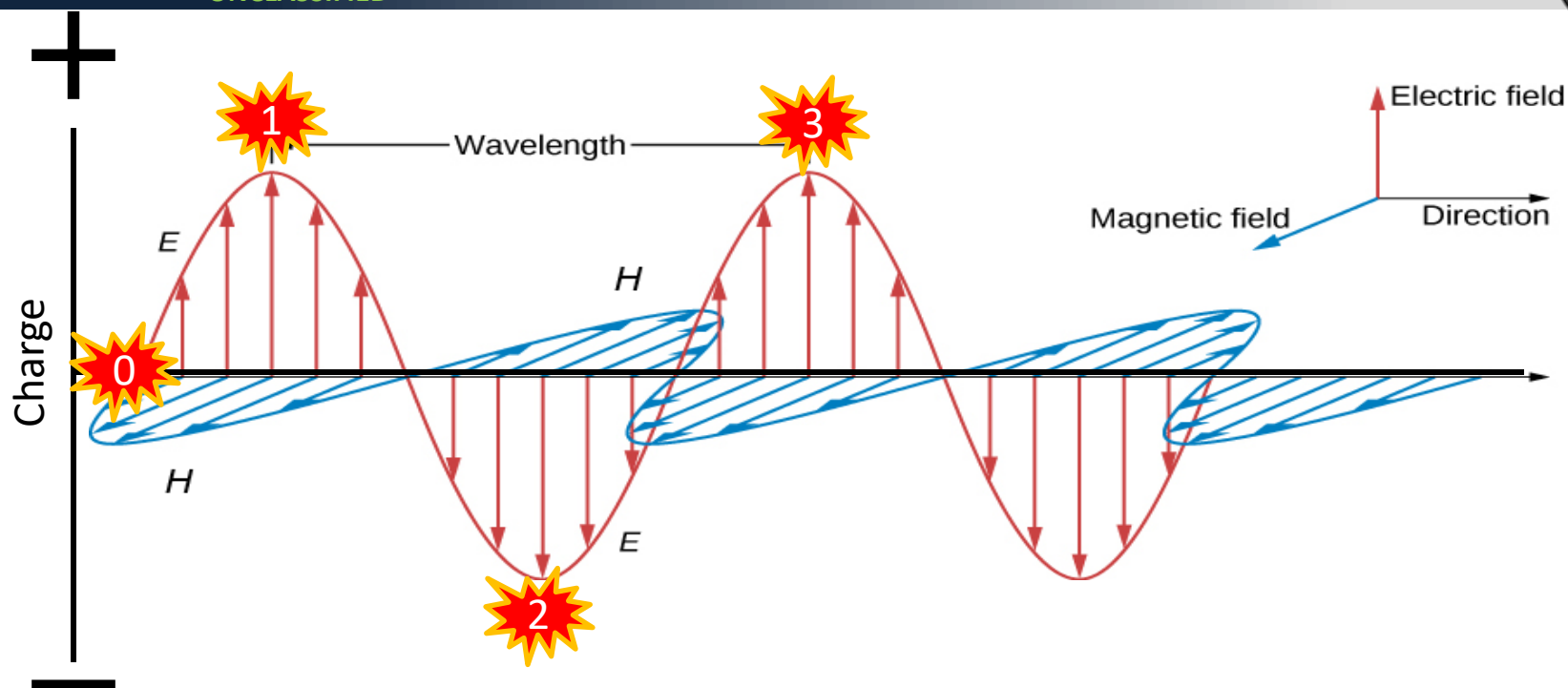
- Normal waves are collections of particles or molecules that push each on each other as force is applied to the medium, they occupy i.e.(water).
- The continuous push motion of the molecules gives the appearance of a wave in the medium.
- If there is a gap in the medium or no medium exists, then particles/molecules can no longer interact, and the wave stops.

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# How EM Waves Move

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- As you move from **0** to **1**, Net change in charge is positive which creates a magnetic field out of the board.
- As you move from **1** to **2**, Net change in charge is negative creating a magnetic field into the board that will begin to create a negative electric field
- As you move from **2** to **3** the net change is once again positive which creates another magnetic field out of the board restarting the cycle

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# Understanding polarization

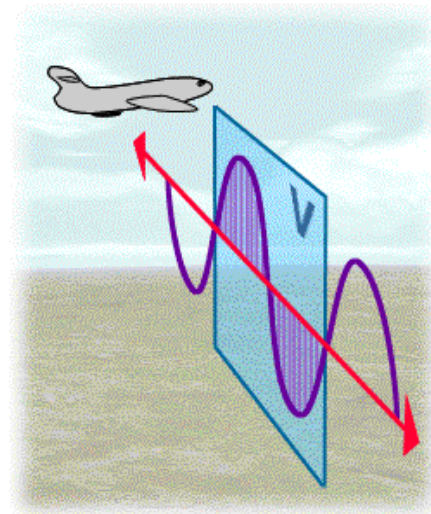
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For any electromagnetic wave:

- The orientation of the electric field determines the polarization of the wave

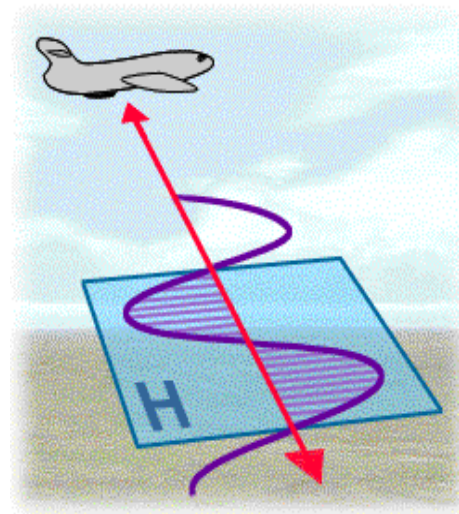
Vertically Polarized:

- Lines of electric force that are at right angles to the Earth's surface, the wave is vertically polarized



Horizontally Polarized:

- Lines of electric force that are parallel to the Earth's surface, the wave is horizontally polarized





# “Right Hand Rule”

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Good rule to determine polarization:

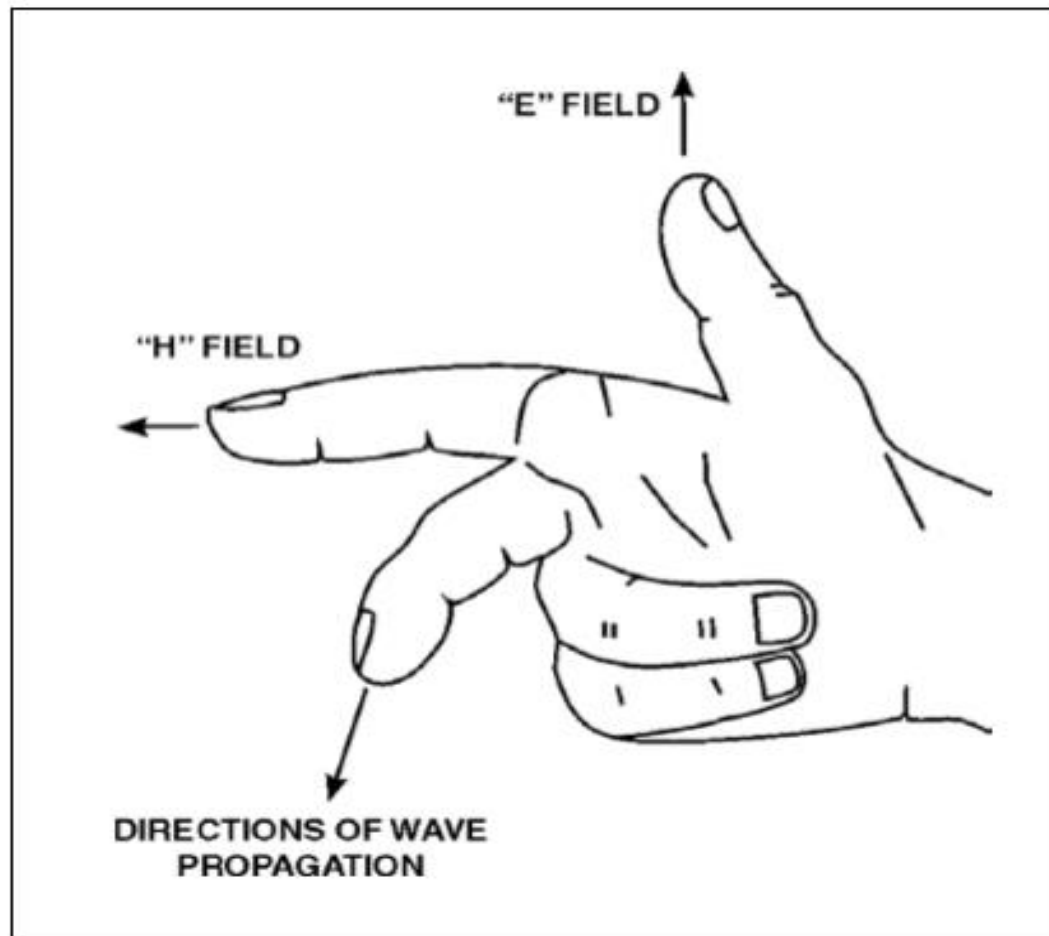
- First: You must know the direction of the electric (E) and magnetic (H) components (your antenna is the electric component)
- Use your right-hand thumb, forefinger and middle finger
- The thumb is the direction of the E field
- The Forefinger is the H Field

Because both the E and H fields reverse directions simultaneously, propagation of a particular wave front is always in the same direction (away from the antenna).



# "Right Hand Rule" Example

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# Application of Polarization

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In wireless communications, antennas must be fully or at least partially oriented in the correct polarization of the wave that it is trying to receive.

- A vertical polarized wave needs a vertical antenna.
- Waves can also be circularly polarized, which combine elements of both horizontal and vertical waves.
- **BLUF: know what antenna you have and the polarization of the targeted signal of interest!**

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# Questions?



# EM Wave Interactions



# Basic EM Wave Interactions

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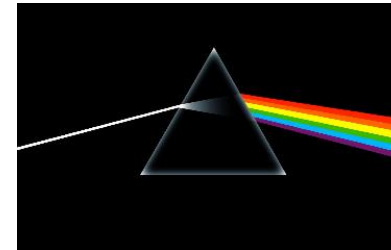
**-Reflection:** The bouncing of waves off of a surface:

- Multi Path Interference
- Ground reflected waves



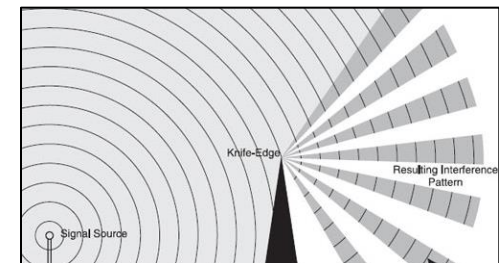
**-Refraction:** The bending of waves as it enters an object:

- Prism
- Tropo Scatter



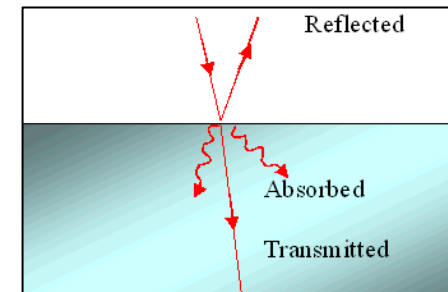
**-Diffraction:** The bending of waves around an object:

- Terrain Masking



**-Absorption:** The loss of energy of a wave as it interacts with an object:

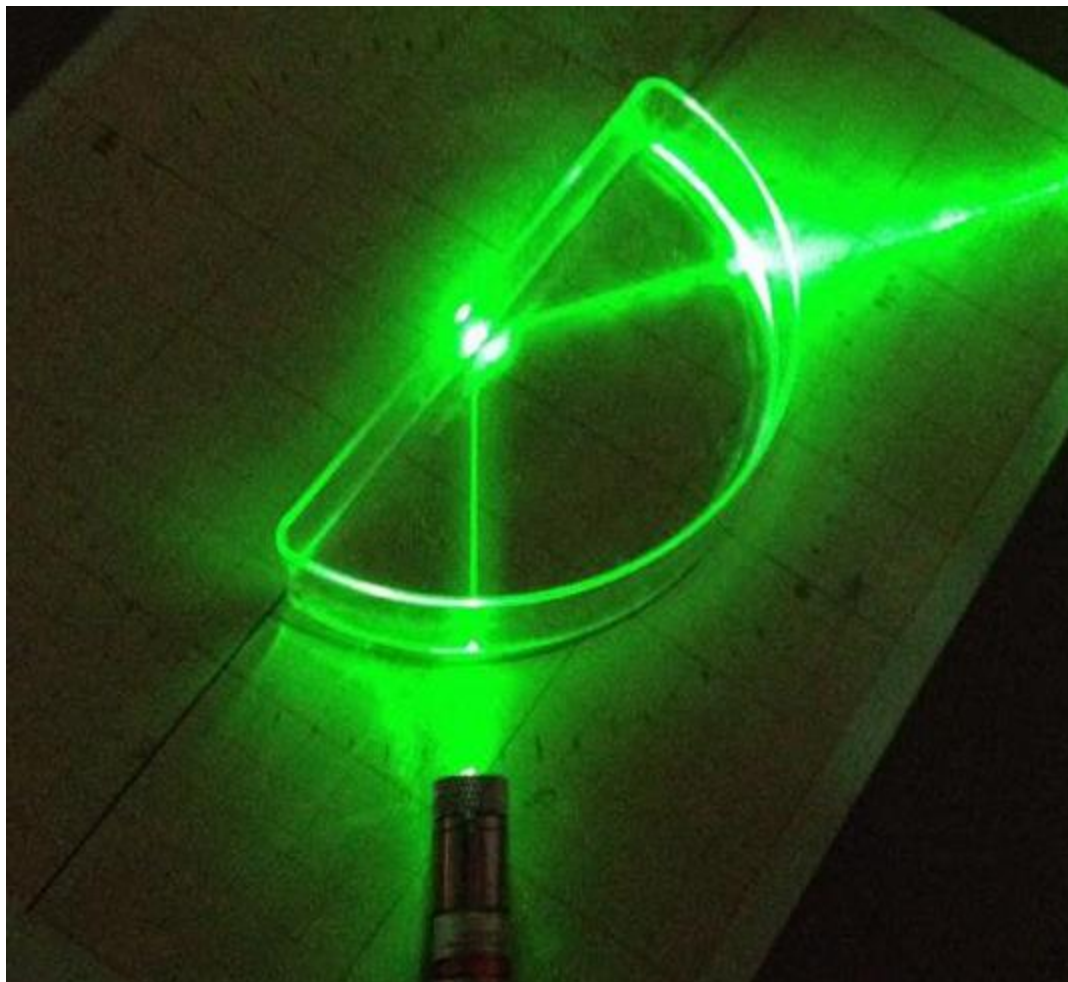
- Signal loss and cloud cover



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# Reflection

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# More on Reflection

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Reflection: When a radio wave hits and bounces off a flat surface (think Earth!!)

- A Phase shift in the wave occurs during reflection (multi path)
- Reflection takes place when waves are  $\sim 180$  degrees out of phase
- Amount of phase shift that occurs is not constant
- Depends on the polarization of the wave and the angle at which the wave strikes the reflecting surface

Radio waves that keep their phase relationships after reflection normally produce a stronger signal

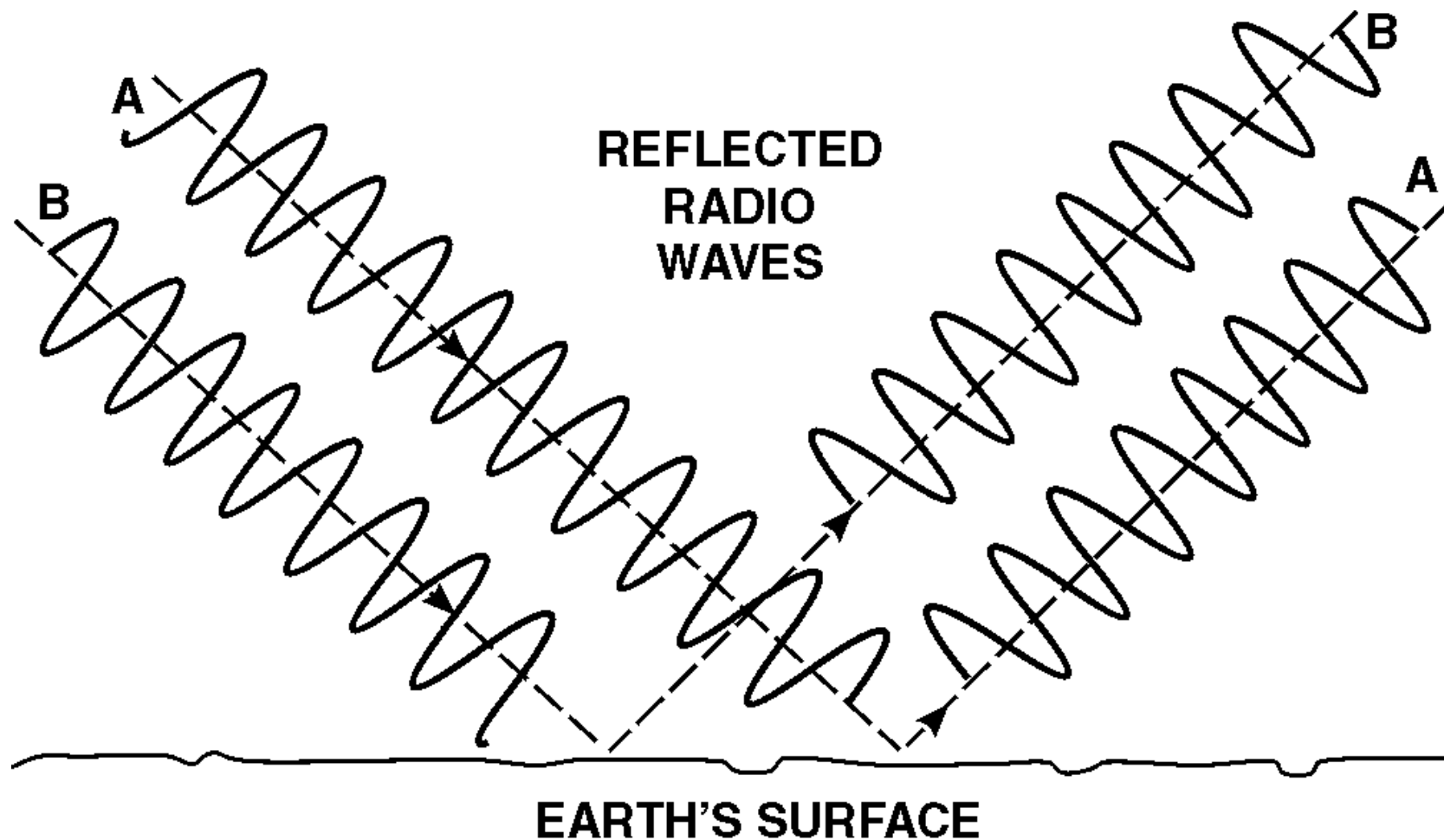
Received out of phase produce a weak or fading signal

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# Phase Shift of Reflected Radio Waves

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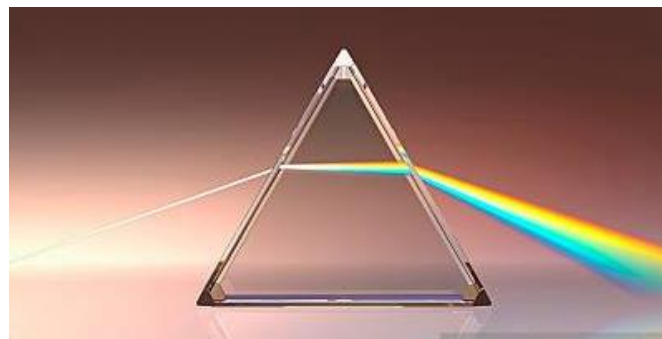
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# Refraction

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Refraction: The bending of a wave as it passes from one medium to another medium of different density

- Wave has to enter at less than 90 degrees
- As it enters, change in speed occurs on one side versus other on wave
- **Multiple applications for beyond Line of Sight communications**



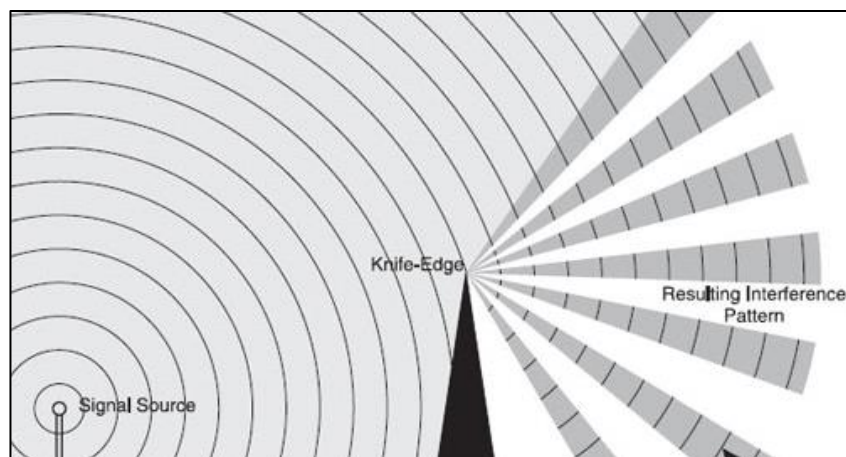
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# Diffraction

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Diffraction:

- Radio waves that meets an obstacle will naturally bend around the obstacle.
- Results in change of direction of PART of the RF wave from line of sight (LOS).
- **Major applications for Electromagnetic Protection (i.e. terrain masking).**



Radio wave passing over either the curved surface of the Earth or a mountain ridge will bend to meet the contours.

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# Absorption

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Absorption: When a radio wave passes through an object or medium some of its energy is absorbed.

Important applications of absorption:

- **Weather can greatly increase the rate of absorption**
- Ground waves lose energy because of ground induced voltage
- Sky wave absorption occurs in high density layers of the ionosphere
- **Absorption is more significant at higher frequencies**

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# Free Space Loss

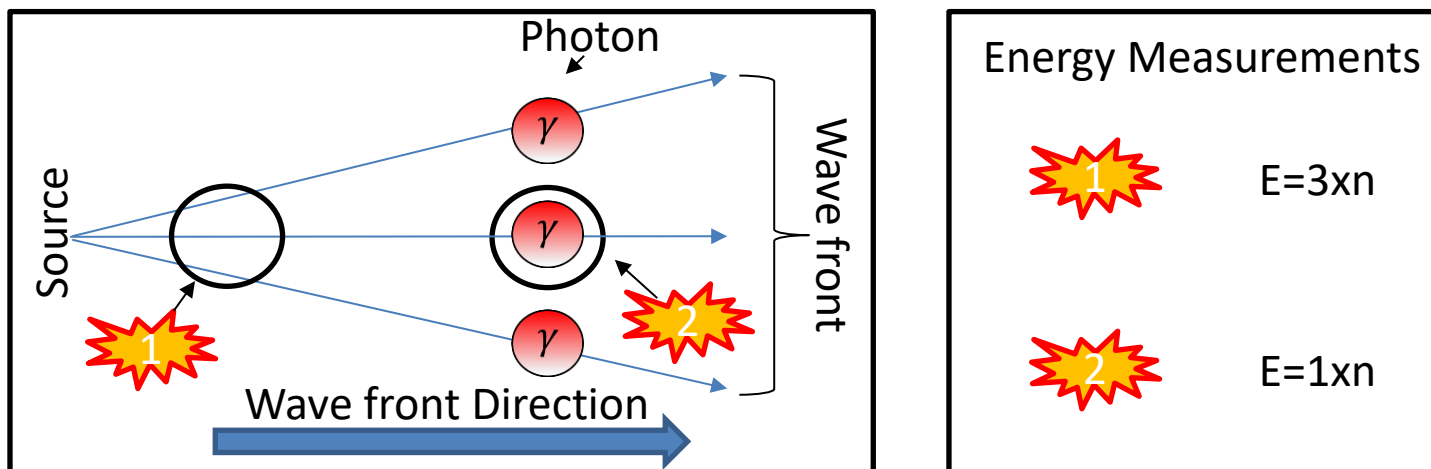
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Free Space Loss is the loss in signal strength that would result from a line-of-sight path through free space (usually air).

Major free space loss is due to the spreading out of the wave front (like a flashlight):

- Energy decreases as distance increases

## Wave Front Spreading



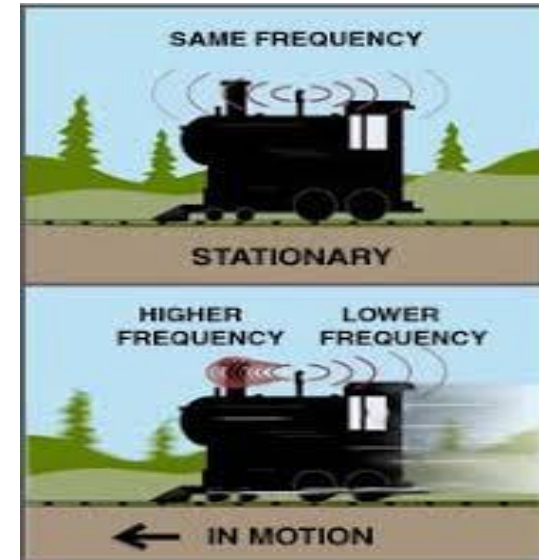


# Doppler Effect in Sound

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**Doppler Effect:** A relative change in frequency or pitch when a sound source moves either toward or away from the listener, or when the listener moves either toward or away from the sound source:

- When the distance between the source and receiver is closing (moving toward each other), the receiver will perceive a higher frequency sound
- When the source and the receiver are moving apart, the receiver will receive a smaller number of sound waves per second and will perceive a lower frequency



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# Doppler Effect in EM Waves

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**EM Doppler Effect:** A relative change in frequency of an EM Wave as the transmitter and receiver move relative to one another.

- The EM Doppler principle is used in equipment such as RADAR, target detection, weapons control, navigation and Electromagnetic Support.
- When an EM wave is moving towards the receiver, it will be measured at a higher frequency. This is called a blue shift.
- When an EM wave is moving away from a receiver it will be measured at a lower frequency. This is called a red shift.



# Application

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Q: Why care about EM waves?

A: Because modern warfare is dependent on C4ISR systems, and those systems use EM waves!

Q: Why care so much about Wave interactions and propagation?

A: Because you live on Earth and there is a lot of obstacles between you and your target!

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# Summary

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- Define radio waves
- Define magnetic, electric, electromagnetic, and radiation fields
- Recognize the characteristics of a radio wave
- Calculate radio wave measurements
- Define how radio waves are polarized
- Define polarization
- Describe different EM wave interactions

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# Questions?